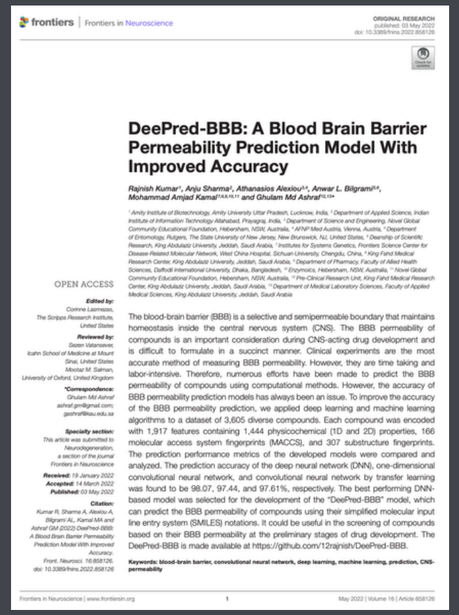
BBB PERMEABILITY MATERIAL AND METHODS

Presented By: Brenden Park

1. Data Collection

Training Dataset



frontiers | Frontiers in Neuroscience

ORIGINAL RESEARCH
published: 03 May 2022
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DeePred-BBB: A Blood Brain Barrier Permeability Prediction Model With Improved Accuracy

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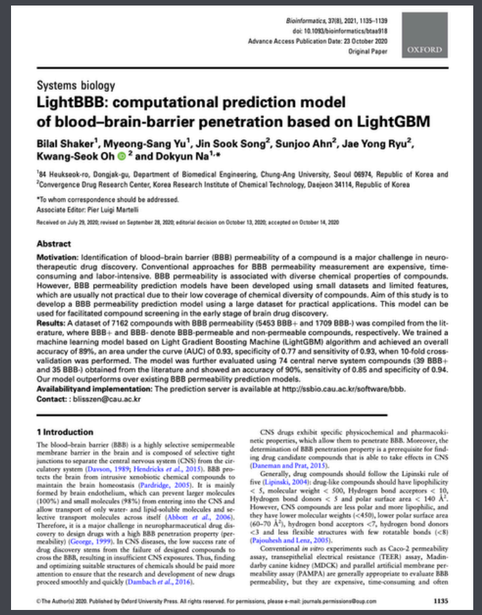
The blood-brain barrier (BBB) is a selective and semipermeable boundary that maintains homeostasis inside the central nervous system (CNS). The BBB permeability of compounds is an important consideration during CNS-acting drug development and is difficult to formulate in a succinct manner. Clinical experiments are the most accurate method of measuring BBB permeability. However, they are time taking and labor-intensive. Therefore, numerous efforts have been made to predict the BBB permeability of compounds using computational methods. However, the accuracy of BBB permeability prediction models has always been an issue. To improve the accuracy of the BBB permeability prediction, we applied deep learning and machine learning algorithms to a dataset of 3,605 diverse compounds. Each compound was encoded with 1,917 features containing 1,444 physicochemical (1D and 2D) properties, 166 molecular access system fingerprints (MACCS), and 307 substructure fingerprints. The prediction performance metrics of the developed models were compared and analyzed. The prediction accuracy of the deep neural network (DNN), one-dimensional convolutional neural network, and convolutional neural network by transfer learning was found to be 98.07, 97.44, and 97.61%, respectively. The best performing DNN-based model was selected for the development of the "DeePred-BBB" model, which can predict the BBB permeability of compounds using their simplified molecular input line entry system (SMILES) notations. It could be useful in the screening of compounds based on their BBB permeability at the preliminary stages of drug development. The DeePred-BBB is made available at https://github.com/12rajnish/DeePred-BBB.

Keywords: blood-brain barrier, convolutional neural network, deep learning, machine learning, prediction, CNS-permeability

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DeePred-BBB

3605 compounds



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Advance Access Publication Date: 23 October 2020
Original Paper

OXFORD

Systems biology

LightBBB: computational prediction model of blood–brain-barrier penetration based on LightGBM

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Associate Editor: Pier Luigi Martelli

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Abstract

Motivation: Identification of blood-brain barrier (BBB) permeability of a compound is a major challenge in neuro-therapeutic drug discovery. Conventional approaches for BBB permeability measurement are expensive, time-consuming and labor-intensive. BBB permeability is associated with diverse chemical properties of compounds. However, BBB permeability prediction models have been developed using small datasets and limited features, which are usually not practical due to their low coverage of chemical diversity of compounds. Aim of this study is to develop a BBB permeability prediction model using a large dataset for practical applications. This model can be used for facilitated compound screening in the early stage of brain drug discovery.

Results: A dataset of 7162 compounds with BBB permeability (5453 BBB+ and 1709 BBB-) was compiled from the literature, where BBB+ and BBB- denote BBB-permeable and non-permeable compounds, respectively. We trained a machine learning model based on Light Gradient Boosting Machine (LightGBM) algorithm and achieved an overall accuracy of 89%, an area under the curve (AUC) of 0.93, specificity of 0.77 and sensitivity of 0.93, when 10-fold cross-validation was performed. The model was further evaluated using 74 central nerve system compounds (39 BBB+ and 35 BBB-) obtained from the literature and showed an accuracy of 90%, sensitivity of 0.85 and specificity of 0.94. Our model outperforms over existing BBB permeability prediction models.

Availabilityand implementation: The prediction server is available at http://ssbio.cau.ac.kr/software/bbb.

Contact: : blisszen@cau.ac.kr

1 Introduction

The blood–brain barrier (BBB) is a highly selective semipermeable membrane barrier in the brain and is composed of selective tight junctions to separate the central nervous system (CNS) from the circulatory system (Davson, 1989; Hendrickx *et al.*, 2015). BBB protects the brain from intrusive xenobiotic chemical compounds to maintain the brain homeostasis (Pardridge, 2005). It is mainly formed by brain endothelium, which can prevent larger molecules (100%) and small molecules (98%) from entering into the CNS and allow transport of only water- and lipid-soluble molecules and selective transport molecules across itself (Abbott *et al.*, 2006). Therefore, it is a major challenge in neuropharmaceutical drug discovery to design drugs with a high BBB penetration property (permeability) (George, 1999). In CNS diseases, the low success rate of drug discovery stems from the failure of designed compounds to cross the BBB, resulting in insufficient CNS exposures. Thus, finding and optimizing suitable structures of chemicals should be paid more attention to ensure that the research and development of new drugs proceed smoothly and quickly (Dambach *et al.*, 2016).

CNS drugs exhibit specific physicochemical and pharmacokinetic properties, which allow them to penetrate BBB. Moreover, the determination of BBB penetration property is a prerequisite for finding drug candidate compounds that is able to take effects in CNS (Banerman and Pratt, 2015).

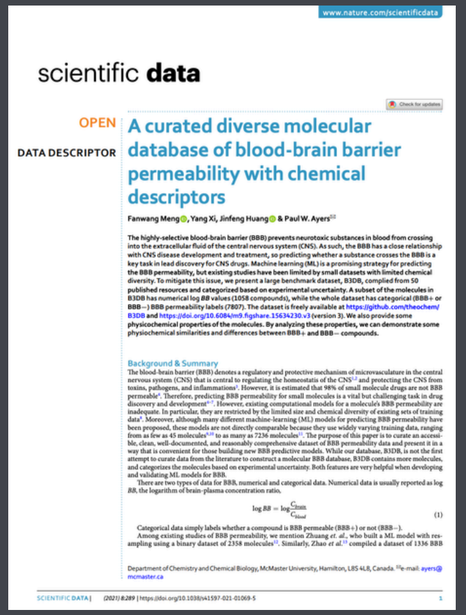
Generally, drug compounds should follow the Lipinski rule of five (Lipinski, 2004): drug-like compounds should have lipophilicity < 5, molecular weight < 500, Hydrogen bond acceptors < 10, Hydrogen bond donors < 5 and polar surface area < 140 Å^2. However, CNS compounds are less polar and more lipophilic, and they have lower molecular weights (<450), lower polar surface area (60–70 Å^2), hydrogen bond acceptors <7, hydrogen bond donors <3 and less flexible structures with few rotatable bonds (<8) (Pajouhesh and Lenz, 2005).

Conventional *in vitro* experiments such as Caco-2 permeability assay, transepithelial electrical resistance (TEER) assay, Madin-darby canine kidney (MDCK) and parallel artificial membrane permeability assay (PAMPA) are generally appropriate to evaluate BBB permeability, but they are expensive, time-consuming and often

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LightBBB

7162 compounds



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DATA DESCRIPTOR

A curated diverse molecular database of blood-brain barrier permeability with chemical descriptors

Fanwang Meng, Yang Xi, Jinfeng Huang & Paul W. Ayers

The highly-selective blood-brain barrier (BBB) prevents neurotoxic substances in blood from crossing into the extracellular fluid of the central nervous system (CNS). As such, the BBB has a close relationship with CNS disease development and treatment, so predicting whether a substance crosses the BBB is a key task in lead discovery for CNS drugs. Machine learning (ML) is a promising strategy for predicting the BBB permeability, but existing studies have been limited by small datasets with limited chemical diversity. To mitigate this issue, we present a large benchmark dataset, B3DB, compiled from 50 published resources and categorized based on experimental uncertainty. A subset of the molecules in B3DB has numerical log *BB* values (1058 compounds), while the whole dataset has categorical (BBB+ or BBB−) BBB permeability labels (7807). The dataset is freely available at https://github.com/theochem/B3DB and https://doi.org/10.6084/m9.figshare.15634230.v3 (version 3). We also provide some physicochemical properties of the molecules. By analyzing these properties, we can demonstrate some physicochemical similarities and differences between BBB+ and BBB− compounds.

Background & Summary

The blood–brain barrier (BBB) denotes a regulatory and protective mechanism of microvasculature in the central nervous system (CNS) that is central to regulating the homeostasis of the CNS[1,2] and protecting the CNS from toxins, pathogens, and inflammations[3]. However, it is estimated that 98% of small molecule drugs are not BBB permeable[4]. Therefore, predicting BBB permeability for small molecules is a vital but challenging task in drug discovery and development[4–7]. However, existing computational models for a molecule's BBB permeability are inadequate. In particular, they are restricted by the limited size and chemical diversity of existing sets of training data[8]. Moreover, although many different machine learning (ML) models for predicting BBB permeability have been proposed, these models are not directly comparable because they use widely varying training data, ranging from as few as 45 molecules[9,10] to as many as 7236 molecules[11]. The purpose of this paper is to curate an accessible, clean, well-documented, and reasonably comprehensive dataset of BBB permeability data and present it in a way that is convenient for those building new BBB predictive models. While our database, B3DB, is not the first attempt to curate data from the literature to construct a molecular BBB database, B3DB contains more molecules, and categorizes the molecules based on experimental uncertainty. Both features are very helpful when developing and validating ML models for BBB.

There are two types of data for BBB, numerical and categorical data. Numerical data is usually reported as log *BB*, the logarithm of brain-plasma concentration ratio.

$$\log BB = \log \frac{C_{\text{brain}}}{C_{\text{blood}}} \quad (1)$$

Categorical data simply labels whether a compound is BBB permeable (BBB+) or not (BBB−).

Among existing studies of BBB permeability, we mention Zhuang *et. al.*, who built a ML model with resampling using a binary dataset of 2358 molecules[12]. Similarly, Zhao *et al.*[13] compiled a dataset of 1336 BBB

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B3DB

1058 compounds



RESEARCH ARTICLE Open Access

Blood–brain barrier penetration prediction enhanced by uncertainty estimation

Xiaochu Tong[1,2], Dingyan Wang[1,2], Xiaoyu Ding[1,2], Xiaoqin Tan[1,2], Qun Ren[3,1], Geng Chen[1,2,4], Yu Rong[5], Tingyang Xu[5], Junzhou Huang[5], Hualiang Jiang[1,2], Mingyue Zheng[1,2*] and Xutong Li[1,2*]

Abstract

Blood–brain barrier is a pivotal factor to be considered in the process of central nervous system (CNS) drug development, and it is of great significance to rapidly explore the blood–brain barrier permeability (BBBp) of compounds in silico in early drug discovery process. Here, we focus on whether and how uncertainty estimation methods improve in silico BBBp models. We briefly surveyed the current state of in silico BBBp prediction and uncertainty estimation methods of deep learning models, and curated an independent dataset to determine the reliability of the state-of-the-art algorithms. The results exhibit that, despite the comparable performance on BBBp prediction between graph neural networks-based deep learning models and conventional physicochemical-based machine learning models, the GROVER-BBBp model shows greatly improvement when using uncertainty estimations. In particular, the strategy combined Entropy and MC-dropout can increase the accuracy of distinguishing BBB + from BBB − to above 99% by extracting predictions with high confidence level (uncertainty score < 0.1). Case studies on preclinical/clinical drugs for Alzheimer's disease and marketed antitumor drugs that verified by literature proved the application value of uncertainty estimation enhanced BBBp prediction model, that may facilitate the drug discovery in the field of CNS diseases and metastatic brain tumors.

Keywords: Blood–brain barrier penetration, BBBp prediction, Uncertainty estimation

Introduction

With the development of society and the aging of the population, central nervous system (CNS) diseases have become the second largest disease after cardiovascular diseases. However, the success rate of clinical candidate CNS drugs is only about 8%, which is quite low compared with the success rate of 20% for cardiovascular diseases [1]. Cancer is also a major disease in the current society, and the success rate of clinical candidates is only about 5% [2]. Worse still, brain metastases are a common route of disease progression in 20% of patients with cancer. The vast majority of patients with brain metastases have a poor prognosis even with the treatment of whole-brain radiation therapy. Second-generation kinase inhibitor with blood–brain barrier (BBB) permeability is believed to be one of the effective treatments of brain metastases [3, 4].

Many potential drugs have been discontinued during their development for clinical use for their insufficient quantity to the CNS because the presence of a BBB. BBB is formed by the endothelial cells of the brain capillaries, which controls the transport of molecules between central nervous system and circulatory system, protects the brain from the damage of toxic compounds and maintains the homeostasis inside the CNS [5]. BBB permeability (BBBp) of compounds is affected by many mechanisms. In the clinical applications of CNS drugs, small lipophilic molecules can cross plasmatic membranes to enter the

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BMC

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MoleculeNet

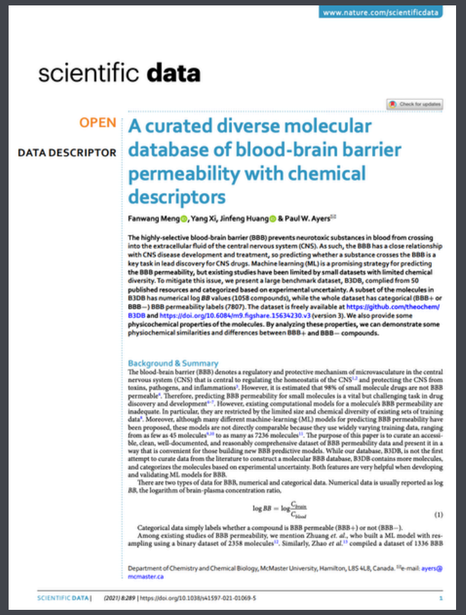
1937 compounds

Training: 13762 compounds before curation

(9866 BBB+, 3296 BBB−)

1. Data Collection

Testing Dataset



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DATA DESCRIPTOR

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Fanwang Meng, Yang Xi, Jinfeng Huang & Paul W. Ayers

The highly-selective blood-brain barrier (BBB) prevents neurotoxic substances in blood from crossing into the extracellular fluid of the central nervous system (CNS). As such, the BBB has a close relationship with CNS disease development and treatment, so predicting whether a substance crosses the BBB is a key task in lead discovery for CNS drugs. Machine learning (ML) is a promising strategy for predicting the BBB permeability, but existing studies have been limited by small datasets with limited chemical diversity. To mitigate this issue, we present a large benchmark dataset, B3DB, complied from 50 published resources and categorized based on experimental uncertainty. A subset of the molecules in B3DB has numerical log *BB* values (1058 compounds), while the whole dataset has categorical (BBB+ or BBB−) BBB permeability labels (7807). The dataset is freely available at https://github.com/theochem/B3DB and https://doi.org/10.6084/m9.figshare.15634230.v3 (version 3). We also provide some physicochemical properties of the molecules. By analyzing these properties, we can demonstrate some physicochemical similarities and differences between BBB+ and BBB− compounds.

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SCIENTIFIC DATA | (2021) 8:289 | https://doi.org/10.1038/s41597-021-01069-5 1

TITAN-BBB

942 compounds

Testing: 942 compounds before curation (638 BBB+, 304 BBB−)

2. Data Curation

1. logBB > 0.3 is equal to BBB+, logBB < -1 is equal to BBB-
2. Remove all invalid SMILES strings (58)
3. Convert SMILES to canonical forms
4. Convert SMILES to isomeric forms (3D arrangement)
5. Remove salts and solvents (111), explicit hydrogens, (37) and duplicate fragments (4)
6. Neutralize non-neutral SMILES (132)
7. Remove duplicate SMILES (7471), conflicting SMILES (250), and any overlapping SMILES in testing dataset (530)

Training: 5378 compounds after curation (3740 BBB+, 1638 BBB-)
Testing: 412 compounds after curation (223 BBB+, 189 BBB-)

3. Model Selection

SMILES

arXiv:2209.01712v1 [cs.LG] 5 Sep 2022

ChemBERTa-2: Towards Chemical Foundation Models

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Abstract

Large pretrained models such as GPT-3 have had tremendous impact on modern natural language processing by leveraging self-supervised learning to learn salient representations that can be used to readily finetune on a wide variety of downstream tasks [1]. We investigate the possibility of transferring such advances to *molecular* machine learning by building a chemical foundation model, ChemBERTa-2, using the "language" of SMILES. While labeled data for molecular prediction tasks is typically scarce, libraries of SMILES strings are readily available.

In this work, we build upon ChemBERTa [2] by optimizing the pretraining process. We compare multi-task and self-supervised pretraining by varying hyperparameters and pretraining dataset size, up to 77M compounds from PubChem. To our knowledge, the 77M set constitutes one of the largest datasets used for molecular pretraining to date. We find that with these pretraining improvements, we are competitive with existing state-of-the-art architectures on the MoleculeNet [3] benchmark suite. We analyze the degree to which improvements in pretraining translate to improvement on downstream tasks.

1 Motivation

Over the past few years, transformers [4, 5] have emerged as popular architectures for learning self-supervised representations of molecules from text representations. ChemBERTa [2] introduced a BERT-like transformer model that learns molecular fingerprints through semi-supervised pretraining and pretrained it on a dataset of 10M compounds. MolBERT [6] experiments with a number of different pretraining objectives on a dataset of 1.6M compounds. SMILES-BERT [7] pretrains on 18.7M compounds from Zinc.

ChemBERTa-2 is a BERT-like transformer model [8] that learns molecular fingerprints through semi-supervised pretraining of the language model. ChemBERTa-2 employs masked-language modelling (MLM) and multi-task regression (MTR) over a large corpus of 77 million SMILES strings, a well-known text representation of molecules. SMILES, is its own language, with a simple vocabulary, consisting of a series of characters representing atom and bond symbols, and very few grammar rules. [9]. ChemBERTa-2 explores the scaling hypothesis that pretraining effectively on larger datasets can yield improved performance, using the largest training dataset in molecular representation learning.

*Equal contribution

2021 ELLIS Machine Learning for Molecule Discovery Workshop.

ChemBERTa-2

arXiv:2211.16349v1 [cs.LG] 29 Nov 2022

BARTSmiles: Generative Masked Language Models for Molecular Representations

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Abstract

We discover a robust self-supervised strategy tailored towards molecular representations for generative masked language models through a series of tailored, in-depth ablations. Using this pre-training strategy, we train `BARTSmiles`, a BART-like model with an order of magnitude more compute than previous self-supervised molecular representations. In-depth evaluations show that `BARTSmiles` consistently outperforms other self-supervised representations across classification, regression, and generation tasks setting a new state-of-the-art on 11 tasks. We then quantitatively show that when applied to the molecular domain, the BART objective learns representations that implicitly encode our downstream tasks of interest. For example, by selecting seven neurons from a frozen `BARTSmiles`, we can obtain a model having performance within two percentage points of the full fine-tuned model on task Clintox. Lastly, we show that standard attribution interpretability methods, when applied to `BARTSmiles`, highlight certain substructures that chemists use to explain specific properties of molecules. The code and the pretrained model are publicly available.

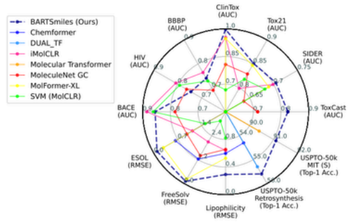


Figure 1: Performance of `BARTSmiles` on a variety of tasks compared to the current state-of-the-art.

*Joint First Authors. Correspondence: armenag@meta.com

BARTSmiles

arXiv:2106.09553v3 [cs.LG] 14 Dec 2022

Large-Scale Chemical Language Representations Capture Molecular Structure and Properties

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ABSTRACT

Predicting the properties of a chemical molecule is of great importance in many applications, including drug discovery and material design. Machine learning-based models promise to enable more accurate and faster molecular property predictions than the current state-of-the-art techniques, such as Density Functional Theory calculations or wet-lab experiments. Various supervised machine learning models, including graph neural nets, have demonstrated promising performance in molecular property prediction tasks. However, the vast chemical space and the limited availability of property labels make supervised learning challenging, calling for learning a general-purpose molecular representation. Recently, unsupervised transformer-based language models pre-trained on large unlabeled corpus have produced state-of-the-art results in many downstream natural language processing tasks. Inspired by this development, we present molecular embeddings obtained by training an efficient transformer encoder model, MoLFormer, which uses rotary positional embeddings. This model employs a linear attention mechanism, coupled with highly distributed training, on SMILES sequences of 1.1 billion unlabeled molecules from the PubChem and ZINC datasets. Experiments show that utilizing the learned molecular representation outperforms existing baselines on downstream tasks, including supervised and self-supervised graph neural net baselines and language models, on several classification and regression tasks from ten benchmark datasets while performing competitively on two others. Further analyses, specifically through the lens of attention, demonstrate that MoLFormer trained on chemical SMILES indeed learns the spatial relationships between atoms within a molecule. These results provide encouraging evidence that the large-scale molecular language models can capture sufficient chemical and structural information to predict various distinct molecular properties, including quantum-chemical properties.

Main

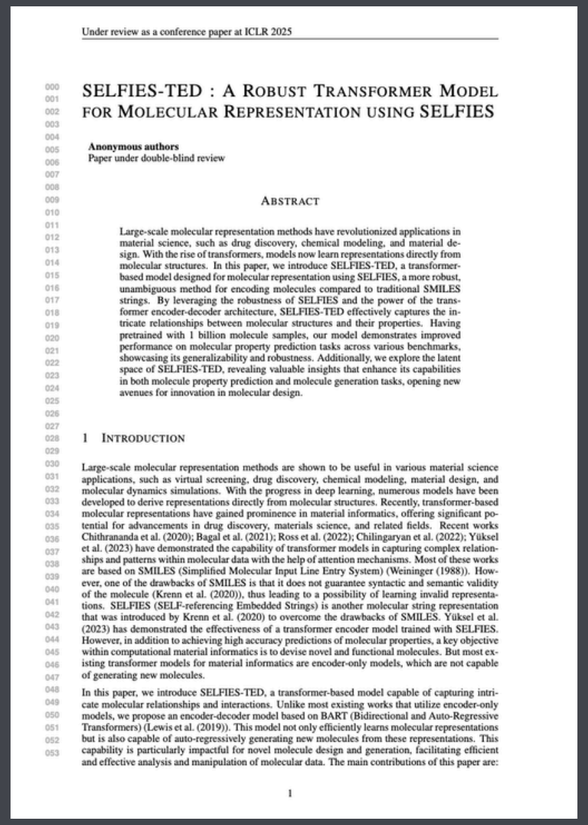
Machine Learning (ML) has emerged as an appealing, computationally efficient approach for predicting molecular properties, with implications in drug discovery and material engineering. ML models for molecules can be trained directly on pre-defined chemical descriptors, such as unsupervised molecular fingerprints[1], or hand-derived derivatives of geometric features such as a Coulomb Matrix (CM)[2]. However, more recent ML models have focused on automatically learning the features either from the natural graphs that encode the connectivity information or from the line annotations of molecular structures, such as the popular SMILES[3] (Simplified Molecular-Input Line Entry System) representation. SMILES defines a character string representation of a molecule by performing a depth-first pre-order spanning tree traversal of the molecular graph, generating symbols for each atom, bond, tree-traversal decision, and broken cycles. Therefore, the resulting character string corresponds to a flattening of a spanning tree of the molecular graph. Learning on SMILES has been widely adopted for molecular property prediction[4–7] as SMILES is generally more compact than other methods of representing structure, including graphs. Additionally, meaningful substructures such as branches, cyclic structures, and chirality information are explicitly represented in SMILES strings, which is not the case for the graph representation.

However, the SMILES grammar is complex and restrictive; most sequences over the appropriate character set do not belong to well-defined molecules. Alternative string-based representations exist, such as SMARTS[8] and SELFIES[9]. Comparing benefits of these alternative representations with respect to SMILES is an active area of research. For example[10], focusing on molecular optimization tasks on the learned representation space, suggested no obvious shortcoming of SMILES with respect to SELFIES in terms of optimization ability and sample efficiency, particularly when the language model is more advanced. Nevertheless, string-based representations are thought to not be topologically-aware, while graphs are. Due to these limitations, deep chemical language models may focus on learning the grammar of molecular strings and not the implicit topological structure of the molecular graphs. Accordingly, while string-based deep neural nets have been employed in predicting molecular properties[5–7,11], they are typically outperformed by graph neural networks (GNNs)[12] and their variants[13–21]. GNN frameworks can be generally viewed as "message passing", which includes local neighborhood information aggregation and information

MoLFormer

3. Model Selection

SELFIES



Under review as a conference paper at ICLR 2025

SELFIES-TED : A ROBUST TRANSFORMER MODEL FOR MOLECULAR REPRESENTATION USING SELFIES

Anonymous authors
Paper under double-blind review

ABSTRACT

Large-scale molecular representation methods have revolutionized applications in material science, such as drug discovery, chemical modeling, and material design. With the rise of transformers, models now learn representations directly from molecular structures. In this paper, we introduce SELFIES-TED, a transformer-based model designed for molecular representation using SELFIES, a more robust, unambiguous method for encoding molecules compared to traditional SMILES strings. By leveraging the robustness of SELFIES and the power of the transformer encoder-decoder architecture, SELFIES-TED effectively captures the intricate relationships between molecular structures and their properties. Having pretrained with 1 billion molecule samples, our model demonstrates improved performance on molecular property prediction tasks across various benchmarks, showcasing its generalizability and robustness. Additionally, we explore the latent space of SELFIES-TED, revealing valuable insights that enhance its capabilities in both molecule property prediction and molecule generation tasks, opening new avenues for innovation in molecular design.

1 INTRODUCTION

Large-scale molecular representation methods are shown to be useful in various material science applications, such as virtual screening, drug discovery, chemical modeling, material design, and molecular dynamics simulations. With the progress in deep learning, numerous models have been developed to derive representations directly from molecular structures. Recently, transformer-based molecular representations have gained prominence in material informatics, offering significant potential for advancements in drug discovery, materials science, and related fields. Recent works Chithrananda et al. (2020); Bagal et al. (2021); Ross et al. (2022); Chilingaryan et al. (2022); Yüksel et al. (2023) have demonstrated the capability of transformer models in capturing complex relationships and patterns within molecular data with the help of attention mechanisms. Most of these works are based on SMILES (Simplified Molecular Input Line Entry System) (Weininger (1988)). However, one of the drawbacks of SMILES is that it does not guarantee syntactic and semantic validity of the molecule (Krenn et al. (2020)), thus leading to a possibility of learning invalid representations. SELFIES (SELF-referencing Embedded Strings) is another molecular string representation that was introduced by Krenn et al. (2020) to overcome the drawbacks of SMILES. Yüksel et al. (2023) has demonstrated the effectiveness of a transformer encoder model trained with SELFIES. However, in addition to achieving high accuracy predictions of molecular properties, a key objective within computational material informatics is to devise novel and functional molecules. But most existing transformer models for material informatics are encoder-only models, which are not capable of generating new molecules.

In this paper, we introduce SELFIES-TED, a transformer-based model capable of capturing intricate molecular relationships and interactions. Unlike most existing works that utilize encoder-only models, we propose an encoder-decoder model based on BART (Bidirectional and Auto-Regressive Transformers) (Lewis et al. (2019)). This model not only efficiently learns molecular representations but is also capable of auto-regressively generating new molecules from these representations. This capability is particularly impactful for novel molecule design and generation, facilitating efficient and effective analysis and manipulation of molecular data. The main contributions of this paper are:

1

SELFIES-TED



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DOMAIN-AGNOSTIC MOLECULAR GENERATION WITH CHEMICAL FEEDBACK

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ABSTRACT

The generation of molecules with desired properties has become increasingly popular, revolutionizing the way scientists design molecular structures and providing valuable support for chemical and drug design. However, despite the potential of language models in molecule generation, they face challenges such as generating syntactically or chemically flawed molecules, having narrow domain focus, and struggling to create diverse and feasible molecules due to limited annotated data or external molecular databases. To tackle these challenges, we introduce MOLGEN, a pre-trained molecular language model tailored specifically for molecule generation. Through the reconstruction of over 100 million molecular SELFIES, MOLGEN internalizes structural and grammatical insights. This is further enhanced by domain-agnostic molecular prefix tuning, fostering robust knowledge transfer across diverse domains. Importantly, our chemical feedback paradigm steers the model away from "*molecular hallucinations*", ensuring alignment between the model's estimated probabilities and real-world chemical preferences. Extensive experiments on well-known benchmarks underscore MOLGEN's optimization capabilities in properties such as penalized logP, QED, and molecular docking. Additional analyses confirm its proficiency in accurately capturing molecule distributions, discerning intricate structural patterns, and efficiently exploring the chemical space[1].

1 INTRODUCTION

Molecule generation – synthesizing and designing novel molecules with desirable properties – holds an important place in chemical science, with numerous applications in drug discovery (Wang et al., 2022). Generating molecules is challenging due to the immense and discrete nature of the molecular space, which, with an estimated size of 10^{33}, makes exhaustive searches impractical (Polishchuk et al., 2013). Early, deep generative models (Jin et al., 2020; Zang & Wang, 2020; Luo et al., 2021; Shi et al., 2020b) have emerged as one of the most promising tools for exploring the broader synthetically accessible chemical space. These models' ability to automatically generate chemically valid and structurally similar molecules has proven to be invaluable for tasks such as the inverse design of functional compounds (Flam-Shepherd et al., 2022).

Current deep generative models typically involve initial training of an unconditional generative model through a large set of existing molecules, and then use additional reward functions (Cao & Kipf, 2018; Popova et al., 2018; You et al., 2018; Popova et al., 2019; Shi et al., 2020b; Zang & Wang, 2020) or property predictors (Liu et al., 2018; Jin et al., 2019; Gómez-Bombarelli et al., 2018) to guide the synthesis of new molecules with desired properties. However, these approaches are limited by challenges in training due to the high variance of Reinforcement Learning (RL) (Xie et al., 2021), fixed-dimensional latent generation space (Wang et al., 2023), and expert-provided generation rules (Sun et al., 2022), which impede efficient exploration of the broader chemical space.

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[1]Code is available at https://github.com/zjunlp/MolGen.

1

Mol-Gen

4. Model Hyperparameter



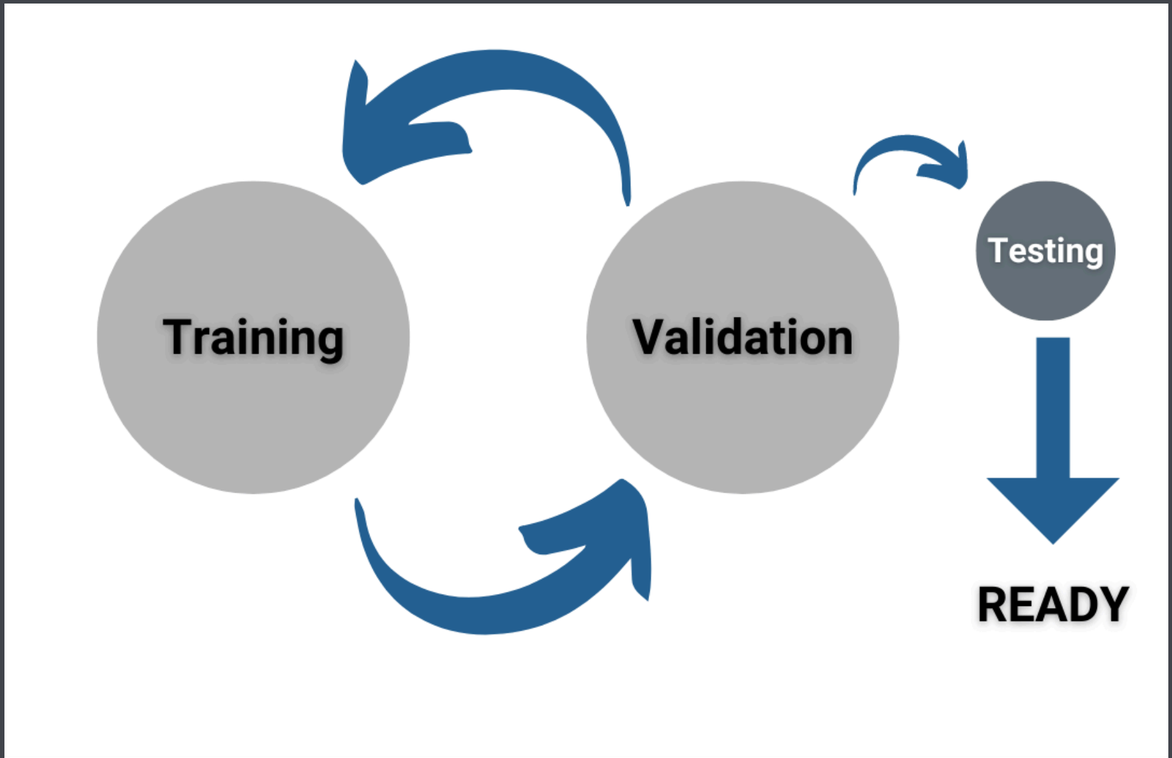
Optuna: An open-source Python framework used to optimize model hyperparameters for training

Hyperparameters

- Learning Rate (1e-5, 1e-1)
- Weight Decay (1e-4, 1e-1)
- Beta 1 (0.9, 0.95)
- Beta 2 (0.98, 0.9999)
- Classifier Dropout (0.0, 0.5)

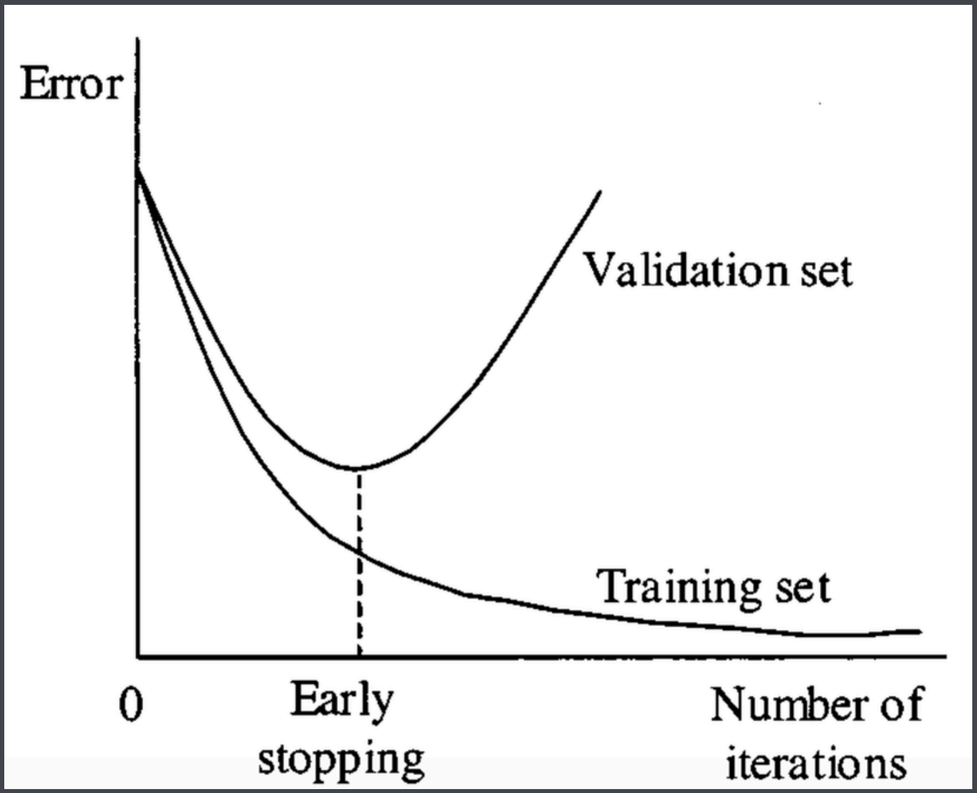
5. Training

Data Split



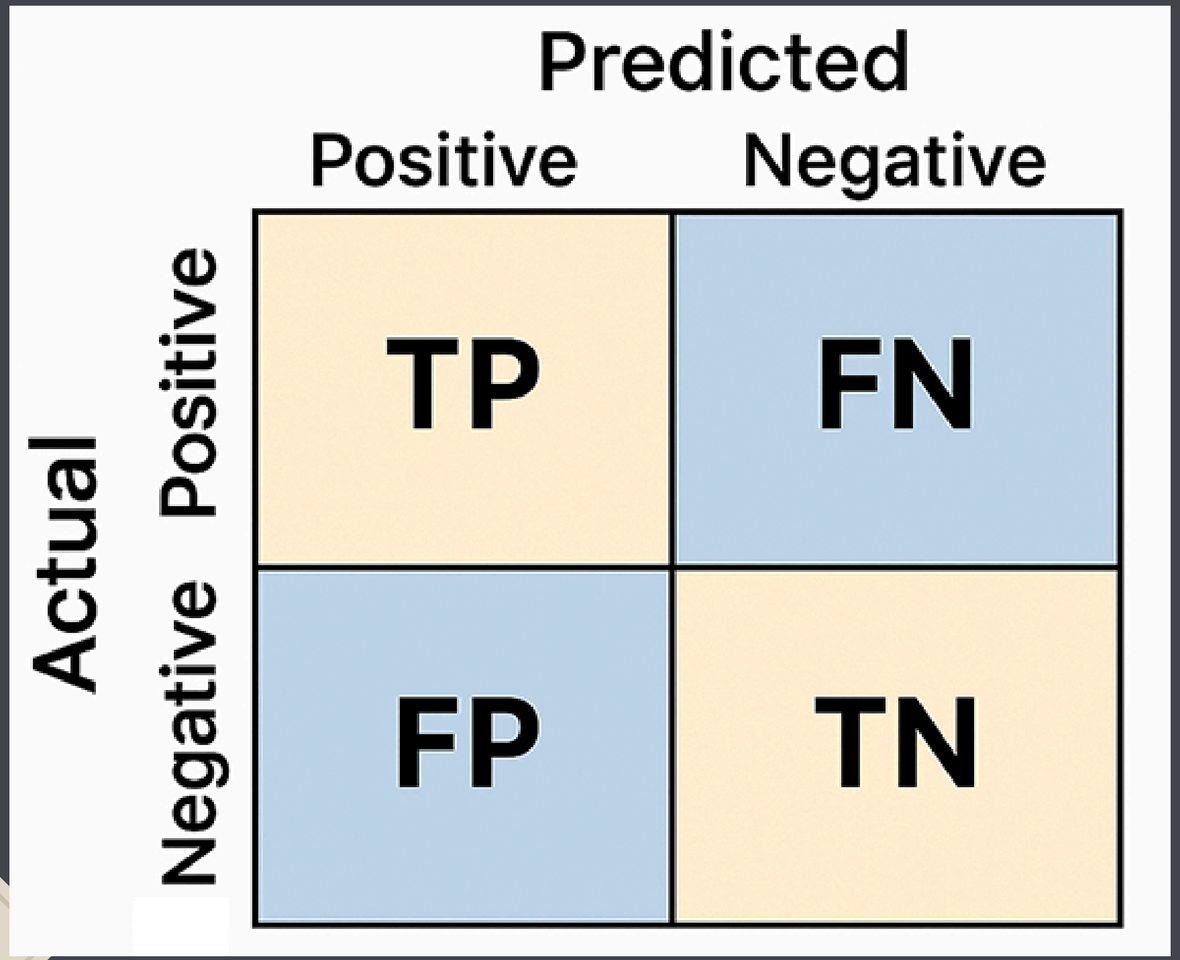
80/20 training/validation ratio with stratification by BBB permeability label

Early Stopping



Train for 50 epochs with early stopping (patience=7) using AdamW

6. Model Evaluation Metric



Confusion Matrix

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall}$$

$$accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

$$specificity = \frac{TN}{TN + FP}$$

Model Metrics

To-Do

1. Complete model training (2/5 currently done) ~ 5 days
2. Research and implement model results graphs into code ~ 5 days

THANK YOU